Primer on Locally Convex Topological Vector Spaces

There are two primary notions we are concerned with when it comes to locally convex topological vector spaces: the notion of a topological vector space, and the notion of local convexity.

Topological Vector Spaces

Definition: Let V be a vector space over $\mathbb{C}$. We say a topology τ on V is compatible with the vector space structure if $(x, y) \mapsto x + y$ and $(\lambda, x) \mapsto \lambda x$ are continuous operations for all $x, y \in V$ and $\lambda \in \mathbb{C}$. If V is equipped with a topology that is compatible with the vector space structure, then V is called a topological vector space (tvs).

Definition: A subset $W \subseteq V$ is said to be *absorbing* if for any $x \in V$, there is some $\lambda_0 \in \mathbb{C}$ such that $x \in \lambda W$ for all $|\lambda| \geq |\lambda_0|$.

We say the subset $C \subseteq V$ is *balanced* if $\lambda C \subseteq C$ for all $|\lambda| \leq 1$.

We say the subset $A \subseteq V$ is *convex* if, for any $x, y \in A$, we have $\lambda x + (1 - \lambda)y \in A$ for all $0 < \lambda < 1$.

Proposition: A topology τ on V is compatible with the vector space structure if and only if τ is translation-invariant and possesses a neighborhood base $\mathcal{B}$ at 0 satisfying

(i) for each $W \in \mathcal{B}$, there is $U \in \mathcal{B}$ such that $U + U \subseteq W$;

(ii) each $W \in \mathcal{B}$ is balanced and absorbing;

(iii) there exists $0 < |\lambda| < 1$ such that whenever $W \in \mathcal{B}$, so too is λW.

Proof. Let V be a tvs. If W is a neighborhood of 0 in V, then there is a neighborhood U of 0 and a real number $\varepsilon > 0$ such that $\lambda U \subseteq W$ whenever $|\lambda| \leq \varepsilon$, by virtue of the continuity of scalar multiplication. Since $\mathbb{C}$ is not discrete, we have

$$O = \bigcup \{\lambda U \mid |\lambda| \leq \varepsilon\}$$

is a balanced neighborhood of 0 that is contained in W. The family $\mathcal{B}$ of all balanced neighborhoods of 0 thus forms a base at 0. Since scalar multiplication is continuous at 0 for every $x_0 \in L$, we have that each $V \in \mathcal{B}$ is absorbing, so we have that (ii) holds. Since vector addition is continuous, we have that $\mathcal{B}$ satisfies condition (i) as well. Finally, for (iii), we observe that for some $0 < |\lambda| < 1$, we have λV for any $V \in \mathcal{B}$ is a balanced neighborhood of 0. Finally, this topology is translation invariant since affine transformations of V are homeomorphisms.

Conversely, let τ be a translation invariant topology on V with a neighborhood base $\mathcal{B}$ at 0 satisfying properties (i) through (iii). We will show that addition and scalar multiplication are continuous operations. We see that $\{x_0 + W \mid W \in \mathcal{B}\}$ is a neighborhood base for $x_0 \in V$, so if we let $U \in \mathcal{B}$ be such that $U + U \subseteq W$, then $x - x_0 \in U$ and $y - y_0 \in U$ implies that $x + y \in x_0 + y_0 + W$, so continuity of vector addition holds.

For continuity of scalar multiplication, let λ_0 and x_0 be fixed elements of $\mathbb{C}$ and V respectively. If we let $W \in \mathcal{B}$, then by (i), there is $U \in \mathcal{B}$ such that $U + U \subseteq W$. Since, from (ii), U is absorbing, there is $\varepsilon > 0$ such that $(\lambda - \lambda_0)x_0 \in U$ whenever $|\lambda - \lambda_0| \leq \varepsilon$. Letting μ satisfy the condition in (iii), there is some $n \in \mathbb{N}$ such that $|\mu^{-n}| \geq |\lambda_0| + \varepsilon$; if we let $O = \mu^n U$, then since U is balanced, if we have $x - x_0 \in W$ and $|\lambda - \lambda_0| \leq \varepsilon$, we have $\lambda(x - x_0) \in U$, and since

$$\lambda x = \lambda_0 x_0 + (\lambda - \lambda_0)x_0 + \lambda(x - x_0),$$

we get $\lambda x \in \lambda_0 x_0 + U + U \subseteq \lambda_0 x_0 + W$. □

Locally Convex Spaces

We start by defining locally convex spaces in a very concrete manner before discussing a characterization in terms of the local base at 0.

Definition: A locally convex topological vector space (or lcs) is a vector space X over $\mathbb{F}$ (either $\mathbb{R}$ or $\mathbb{C}$) paired with a family of seminorms $\mathcal{P}$ that separates the points of X, in the sense that if $p(x) = 0$ for all $p \in \mathcal{P}$, then $x = 0$.

The topology $\tau_{\mathcal{P}}$ is generated by the (necessarily convex) subbasic open sets

$$U(x_0, p, r) = \{x \in X \mid p(x - x_0) < r\}$$

for fixed $x_0 \in X$, $p \in \mathcal{P}$, and $r > 0$.

Theorem: Let X be a lcs.

(i) A neighborhood base at 0 is provided by the convex open sets

$$U(F, r) = \bigcap_{p \in F} \{x \in X \mid p(x) < r\},$$

where $r > 0$ and $F \subseteq \mathcal{P}$ is finite.

(ii) The topology on X is Hausdorff.

(iii) The topology on X is compatible with the vector space structure.

(iv) Convergence of nets in X is given by the following: $(x_i)_i \to x$ if and only if $p(x - x_i) \to 0$ for all $p \in \mathcal{P}$.

Proof.

(i) We see that $\tau_{\mathcal{P}}$ has a base obtained from finite intersections of sets $U(x_i, p_i, r_i)$ for each $1 \leq i \leq n$. Let

$$0 \in \underbrace{\bigcap_{i=1}^{n} U(x_i, p_i, r_i)}_{=:U}$$

and $r = \min\{r_i \mid 1 \leq i \leq n\} - p_i(x_i - 0)$, with $F = \{p_1, \dots, p_n\}$. Then, $0 \in U(F, r) \subseteq U$, since if $x \in U(F, r)$, then

$$\begin{aligned} p_i(x_i - x) &\leq p_i(x_i) + p_i(x) \\ &< p_i(x_i) + r_i - p(x_i) \\ &= r_i, \end{aligned}$$

whence $x \in U$.

(ii) If $x \neq y$, then $x - y \neq 0$, so there is some $p \in \mathcal{P}$ such that $p(x - y) = r > 0$, meaning $U(x, p, r/2)$ and $U(y, p, r/2)$ are disjoint open subsets separating x and y.

(iii) Let $A(x, y) = x + y$ be the addition map, and let $x_0, y_0 \in X$ with $x_0 + y_0 \in U$. Let $F \subseteq \mathcal{P}$ and $r > 0$ be such that $x_0 + y_0 + U(F, r) \subseteq U$. If $x \in x_0 + U(F, r/2)$ and $y \in u_0 + U(F, r/2)$, then for any $p \in F$, we have

$$\begin{aligned} p(x_0 + y_0 - x - y) &\leq p(x_0 - x) + p(y_0 - y) \\ &< \frac{r}{2} + \frac{r}{2} \\ &= r, \end{aligned}$$

Thus, $(x_0 + U(F, r/2)) \times (y_0 + U(F, r/2)) \subseteq A^{-1}(x_0 + y_0 + U(F, r))$. Scalar multiplication works analogously.

(iv) We have that $\lim_{i \in I} x_i = x$ means that, for any finite $F \subseteq \mathcal{P}$ and $r > 0$, there is $i_0 \in I$ such that $x_i \in x + U(F, r)$ for all $i \geq i_0$. If we let $F = \{p\}$, we find that $\lim_{i \in I} p(x - x_i) = 0$ for all $p \in \mathcal{P}$.

Conversely, if $\lim_{i\in I} p(x - x_i) = 0$ for all $p \in \mathcal{P}$, then for some finite $F = \{p_1, \dots, p_n\}$ and $r > 0$, we may chose i_k such that $p_i(x - x_i) < r$ for all $i \geq i_k$. Setting i_0 to be greater than i_k for each $k = 1, \dots, n$, we have that $x_i \in x + U(F, r)$ for each $i \in I$ with $i \geq i_0$.

□

Proposition: Let $f\colon V \to \mathbb{F}$ be a linear map on a lcs. The following are equivalent:

(i) f is continuous;

(ii) f is continuous at 0;

(iii) $\ker(f)$ is closed;

(iv) there is a finite $F = \{p_1, \dots, p_n\} \subseteq \mathcal{P}$ and $t_1, \dots, t_n \in \mathbb{R}_+$ such that

$$|f(x)| \leq \sum_{i=1}^{n} t_i p_i(x)$$

for all $x \in X$.

Proof. We have (i) implies (ii). If (ii) holds, then for some net $(x_i)_i \subseteq \ker(f)$ with $x_i \to x$, we have that $x - x_i \to 0$, so $f(x) - f(x_i) \to 0$ by continuity. Since $f(x_i) = 0$, we have $f(x) = 0$, so $\ker(f)$ is closed, giving (iii).

Now, suppose (iii) holds. Assume that $f \neq 0$, and pick $x_0 \in X$ with $f(x_0) = 1$. Since $\ker(f)$ is closed, there is some neighborhood $x_0 + U(F, r)$ disjoint from $\ker(f)$. We observe that $U(F, r)$ is balanced, as if $|\lambda| \leq 1$ and $x \in U(F, r)$, then

$$\begin{aligned} p_i(\lambda x) &= |\lambda| p_i(x) \\ &\leq p_i(x) \\ &< r, \end{aligned}$$

so that $|f(U(F, r))| \leq 1$. If $\sum_{i=1}^{n} p_i(x) < r$, then $p_i(x) < r$ for each $1 \leq i \leq n$, meaning $|f(x)| < 1$. Scaling gives $|f(x)| \leq \sum_{i=1}^{n} \frac{1}{r} p_i(x)$. This gives (iv).

If (iv) holds, then for some $(x_i)_i \to x$, we have $p_i(x - x_i) \to 0$, meaning

$$\begin{aligned} |f(x) - f(x_i)| &\leq \sum_{i=1}^{n} t_i p_i(x - x_i) \\ &\to 0, \end{aligned}$$

so $f(x_i) \to f(x)$ and f is continuous. □

Elaborating on the remark about $U(F, r)$, we observe that we have that if V is a lcs, then V has a local base at 0 consisting of open, balanced, convex sets.

Geometric Hahn–Banach Theorems and Consequences

There are a variety of theorems related to locally convex topological vector spaces that all fall under the label of "Hahn–Banach," specifically related to separation of subsets by continuous linear functionals (or hyperplanes).

Geometric Hahn–Banach Theorems

Lemma: Let L be a Hausdorff tvs over $\mathbb{R}$ with dimension at least 2. If B is an open, convex subset of L not containing 0, then there exists a one-dimensional subspace of L not intersecting B.

Proof. Let E be any fixed two-dimensional subspace of L. If $E \cap B = \emptyset$, then we are done, so assume that $B_1 = E \cap B$ is nonempty. Then, B_1 is an open, convex subset of E that does not contain 0. We may iden-

tify E with the Euclidean plane; projecting B_1 onto a subset of the unit circle by the map $f\colon \mathbb{R}^2 \setminus \{0\} \to S^1$,

$$(x, y) \mapsto \left(\frac{x}{(x^2+y^2)^{1/2}}, \frac{y}{(x^2+y^2)^{1/2}} \right),$$

we have that $f(B_1)$ is connected as B_1 is connected and f is continuous, and $f(B_1)$ is an open subset of C. Thus, $f(B_1)$ is an open arc on C that subtends an angle less than or equal to π at 0. Consequently, there exists a straight line in E passing through 0 that does not intersect B_1. □

Theorem: Let L be a tvs, and let M be an affine subspace with A a nonempty open convex subset of L not intersecting M. Then, there exists a closed hyperplane in L containing M that does not intersect A.

This is also known as Mazur's theorem, but we can view it as the geometric form of the Hahn–Banach theorem. We recall that a hyperplane is a maximal proper affine subspace.

Proof. After a translation, we may assume that M is a subspace of L. We let $\mathcal{M}$ denote the collection of all closed real subspaces of L that contain M and do not intersect A; since $\overline{M} \in \mathcal{M}$, we have that $\mathcal{M}$ is nonempty.

Ordering $\mathcal{M}$ by inclusion, we observe that $\mathcal{M}$ satisfies the criteria for Zorn's lemma. Thus, there exists a maximal element H_0 of $\mathcal{M}$. If L_0 denotes the real underlying space of L, then L_0/H_0 is Hausdorff, since H_0 is closed. Since $A \neq \emptyset$, we have that L_0/H_0 is (real) dimension at least 1.

Suppose toward contradiction that L_0/H_0 has dimension greater than or equal to 2. Since the quotient map $\pi\colon L_0 \to L_0/H_0$ is linear and open, we have that $B = \pi(A)$ is a convex open subset of L_0/H_0 that does not contain 0 as H_0 does not intersect A. Therefore, there is a one-dimensional subspace N of L_0/H_0 not intersecting B. Then, $\pi^{-1}(N)$ is a closed subspace of L_0 properly containing H_0 not intersecting A, which contradicts the maximality of H_0 in $\mathcal{M}$. Thus, H_0 is a closed real hyperplane containing M and not intersecting A.

In the case where L is a vector space over $\mathbb{C}$, then $M = iM$ when we assume M is a subspace. Consequently, $H_1 = H_0 \cap iH_0$ is a closed hyperplane in L not intersecting A containing M, and we are done. □

By rotating (if necessary) via a factor of $e^{i\theta}$, this result implies that if M is an affine subspace and A is a nonempty open convex subset of L disjoint from M, there is a continuous linear functional $f\colon L \to \mathbb{C}$ and some $t > 0$ such that $A \subseteq \{v \in L \mid \operatorname{Re}(f(v)) > t\}$ and $M \subseteq \{v \in L \mid \operatorname{Re}(f(v)) \leq 0\}$.

Of course, these are all excellent characterizations, but they do not really exist unless there are convex subsets to begin with.

Corollary: Let L be a tvs. Then, there exists a continuous linear functional $f \neq 0$ on L if and only if L contains a proper nonempty open convex subset.

Proof. Suppose $f \neq 0$ is a continuous linear functional on L. Then, the subset $A = \{x \mid |f(x)| < 1\}$ is proper, nonempty, convex and open.

Conversely, if $A \subseteq L$ is open, and $x_0 \notin A$, then x_0 is contained in a closed hyperplane not intersecting A, meaning there is a nonzero continuous linear functional on L. □

This is why most of the characterizations of geometric Hahn–Banach require the hypothesis of the underlying vector space being a lcs.

By including the assumption of our space being a lcs, we may upgrade the separation to be able to separate convex subsets rather than simply convex subsets and affine subspaces.

Theorem: Let V be a lcs, and let A, B be disjoint convex subsets of V. If A is open, then there is a continuous linear functional $f\colon V \to \mathbb{F}$ and $\alpha \in \mathbb{R}$ such that

$$\operatorname{Re}(f(b)) \leq \alpha < \operatorname{Re}(f(a))$$

for all $a \in A$ and $b \in B$.

Proof. Let $U = A - B$. This is an open and convex subset with $0 \notin U$, so there is a hyperplane with $\operatorname{Re}(f(x)) > 0$ for all $x \in U$ (combine the first two results from this subsection). Thus, we have

$$\begin{aligned} 0 &< \operatorname{Re}(f(a-b)) \\ &= \operatorname{Re}(f(a)) - \operatorname{Re}(f(b)), \end{aligned}$$

meaning that $\sup_{b\in B} \operatorname{Re}(f(b)) \coloneqq \alpha \leq \inf_{a\in A} \operatorname{Re}(f(a))$.

We claim that $f(A)$ is open. Let $a \in a + U(F,r) \subseteq A$. If $U(F,r) \subseteq \ker(f)$, then $\ker(f) \supseteq \bigcup_{n\geq 1} U(F,r) = V$, so $f = 0$. Since this is not possible, it follows that $f(U(F,r))$ contains some $z \neq 0$. Since $U(F,r)$ is balanced and convex, so is $f(U(F,r))$, so $f(U(F,r)(0))$ contains an open disc about 0, so $f(A)$ is open, and thus the infimum $\inf_{a\in A} \operatorname{Re}(f(a))$ is not attained. Thus, $\alpha < f(a)$ for all $a \in A$. □

Corollary: Suppose A and B are disjoint open convex subsets of a lcs V. Then, there is a continuous linear functional $f \in V^*$ and $\alpha \in \mathbb{R}$ such that

$$\operatorname{Re}(f(b)) < \alpha < \operatorname{Re}(f(a))$$

for all $a \in A$ and $b \in B$.

Next, we discuss separation of closed convex subsets.

Lemma: Let X be a lcs. If $K \subseteq X$ is compact, V is open, and $K \subseteq V$, then there is a convex open neighborhood U of 0 such that $K + U \subseteq V$.

Proof. For each $x \in K$, there is a basic open neighborhood $x+U(F_x, r_x) \subseteq V$. The family of $x+U(F_x, r_x/2)$ for each $x \in K$ forms an open cover of K, so we may take $x_i + U(F_i, r_i/2)$. If F is the union of the F_i and r is the minimum of the r_i, then if $U = U(F, r/2)$, we have, for any $x \in K$, some x_i such that $x \in x_i + U(F_i, r_i/2)$. We get

$$\begin{aligned} x + U &\subseteq x_i + U(F_i, r_i/2) + U(F, r/2) \\ &\subseteq x_i + U(F_i, r_i/2) + U(F_i, r_i/2) \\ &= x_i + U(F_i, r_i) \\ &\subseteq U. \end{aligned}$$

Thus, $K + U \subseteq V$. □

Corollary: Let X be a lcs, and let A, B be disjoint closed convex subsets of X. If B is compact, then there is $f \in X^*$ and $\beta < \alpha \in \mathbb{R}$ such that

$$\begin{aligned} \sup_{b\in B} \operatorname{Re}(f(b)) &= \beta \\ &< \alpha = \inf_{a\in A} \operatorname{Re}(f(a)). \end{aligned}$$

Proof. There is a convex open set $0 \in U$ such that $B + U \subseteq A^c$, meaning $B + U$ is convex, open, and disjoint from A, so there is $f \in X^*$ and $\alpha \in \mathbb{R}$ such that $\operatorname{Re}(f(b)) < \alpha = \inf_{a\in A} \operatorname{Re}(f(a))$ for all $b \in B + U$. Since B is compact, $\beta = \sup_{b\in B} \operatorname{Re}(f(b))$ is attained, with $\beta < \alpha$. □

Corollary: If X is a lcs, then X^* separates the points of X.

Consequences

Recall that if A is a subset of a vector space X, then the convex hull of A, denoted $\operatorname{conv}(A)$, is the intersection of all convex subsets of X that contain A. The closed convex hull, $\overline{\operatorname{conv}}(A)$, is the same but for the intersection of all closed convex subsets of X that contain A. In fact, there is another way to characterized the closed convex hull if X is a lcs.

Proposition: Let X be a lcs, and let $A \subseteq X$. Then, $\overline{\operatorname{conv}}(A)$ is the intersection of all closed *half-spaces* that contain A, where a half-space is the set $\{x \in X \mid \operatorname{Re}(f(x)) \leq \alpha\}$ for some $f \in X^*$ and $\alpha \in \mathbb{R}$.

It follows that if A is closed and convex, then $x \in A$ if and only if $\operatorname{Re}(f(x)) \leq \sup_{a\in A} \operatorname{Re}(f(a))$ for all

$f \in X^*$.

Proof. We have that $\overline{\operatorname{conv}}(A)$ is contained in the intersection. If $x \notin \overline{\operatorname{conv}}(A)$, then there is $f \in X^*$ such that $\operatorname{Re}(f(x)) > \alpha = \sup_{a \in A} \operatorname{Re}(f(a))$. Thus, this intersection is exactly $\overline{\operatorname{conv}}(A)$. □

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